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This is a summary report of my research assignments I was given by my advisor during my PhD program

SUMMARY REPORT: FOR RESEARCH ASSIGNMENTS OF MY PHD

WEEK 9

Summary:
This week I started 2 courses in Graph theory online.

1. Advanced Algorithmics and Graph Theory with Python, IMTx NET04x.
2. Graph Search, Shortest Paths, and Data Structures, Stanford University.

I have read the MATLAB code for the Voronoi diagram and analysis it to understand the problem, there are some concepts to understand them in graph theory and image processing which I found them in the new courses I am taking online.

I can understand the MATLAB code now much better than before.

I. IMAGE PROCESSING MATLAB COURSE

Assignment: Completed Part III
Completed: Yes
Submitted: NA
Problems: No
Explain: NA
Procedure: Part 3. Segmenting an Image.
Create binary images by thresholding intensity values.

1. Course Example - Photos of Receipts
2. What is Segmentation?
3. Thresholding Intensity Values
4. Working with Binary Images

Part 4. Pre- & Postprocessing Techniques.

1. Improving Segmentation.
2. Filtering Noise.
3. Background Subtraction.
4. Binary Morphology

II. ASSIGNMENT #2 VORONOI DIAGRAM

Assignment: Generate 2 Voronoi diagrams with GVD and 5 figures using MATLAB code provided for the "Voronoi Diagram project memo". Find the paths which was not performed with the Code and find solution for this problem

Completed: No
Submitted: NA
Problems: Previous problem
Procedure: I have read the MATLAB code for the Voronoi diagram and analysis it to understand the problem, there are some concepts to understand them in graph theory and image processing which I found them in the new courses I am taking online. I can understand the MATLAB code now much better than before and I am thinking of applying another algorithm but before that I wanted to make sure where is the problem (planning the path, routing, or the shortest path ..etc)

III. GRAPH SEARCH, SHORTEST PATHS, AND DATA STRUCTURES COURSE

Assignment: Complete Part 1
Completed: No
Submitted: NA
Problems: NA
Explain: NA
Procedure: Part 1. Graph Search And Connectivity

1. Lecture: Graph Search.
2. Lecture Breadth-First Search (BFS): The Basics.

Next Week:

1. Lecture: BFS and Shortest Paths.
2. Lecture: BFS and Undirected Connectivity.
3. Lecture: Depth-First Search (DFS). Lecture Topological Sort.
4. Lecture Computing Strong Components: The Algorithm.

This Report was submitted in 2019 for Mississippi State University, Electrical and Computer Engineering Department during Fall 2019 foe the course of Introduction to Radars Engineering instructed by Dr. Ball John

Meshaal Mouawad, Mississippi State University, Starkville, MS 39759 USA (e-mail: mm4922@msstate.edu).

5. Lecture Computing Strong Components: The Analysis.
6. Lecture Structure of the Web.

IV. IMAGE PROCESSING MATLAB COURSE

Assignment: Completed Part III
Completed: Yes
Submitted: NA
Problems: No
Explain: NA
Procedure: Part 1: Fundamentals of Graph Theory, Problem Solving, Good Programming Practices

1. Addressing a Computational Problem
2. Graphs and Paths
3. Representing Graphs
4. Good programming practices.

Part 2: Graph Traversal, Routing, Queuing Structures :

1. Graph Traversal
2. Routing tables
3. Queuing Structures

Next Week

1. Part 3: Shortest Paths, Min-Heaps, Algorithmic Complexity

1. Dijkstra's Algorithm
2. Min-heaps
3. Algorithm complexity.

Part 4: NP-Completeness, Traveling Salesman Problem, Backtracking

1. Traveling Salesman
2. Bruteforce and Backtracking to solve NP-Complete
3. Problem Complexity and NP-Completeness

V. READING PAPER

Assignment: NA
Completed: NA
Submitted: Na
Problems: Na
Explain: Na
Procedure: Na

Appendix

Advanced Algorithmics and Graph Theory with Python course

The course is made up of 6 sections of roughly 2-4 hours of work each and by the end we'll have gone over data-structures, basic and advanced algorithms for graph theory, complexity/accuracy trade-offs, and even combinatorial game theory. We've prepared a lot of materials for you to improve your knowledge and skills in these topics: videos, quizzes and programming labs!

1. Part 1: Fundamentals of Graph Theory, Problem Solving, Good Programming Practices, Incomplete section
    1. 1. Addressing a Computational Problem, Completed
    2. 2. Graphs and Paths, Completed
    3. 3. Representing Graphs, Completed
    4. 4. Good programming practices, Completed
    5. Quiz 1 (1 Question), Completed

6. Lab 1, Incomplete

2. Part 2: Graph Traversal, Routing, Queuing Structures, Incomplete section
    1. 1. Graph Traversal, Completed
    2. 2. Routing tables, Completed
    3. 3. Queuing Structures, Incomplete
    4. Quiz 2 (1 Question), Incomplete
    5. Lab 2, Incomplete
3. Part 3: Shortest Paths, Min-Heaps, Algorithmic Complexity, Incomplete section
    1. 1. Dijkstra's Algorithm, Incomplete
    2. 2. Min-heaps, Incomplete
    3. 3. Algorithm complexity, Incomplete
    4. Quiz 3 (1 Question), Incomplete
    5. Lab 3, Incomplete
4. Part 4: NP-Completeness, Traveling Salesman Problem, Backtracking, Incomplete section
    1. 1. Traveling Salesman Problem, Incomplete
    2. 2. Bruteforce and Backtracking to solve NP-Complete Problems, Incomplete
    3. 3. Problem Complexity and NP-Completeness, Incomplete
    4. Quiz 4 (1 Question), Incomplete
    5. Lab 4, Incomplete
5. Part 5: Heuristics, Greedy Approaches, Accuracy/Complexity tradeoff, Incomplete section
    1. 1. Heuristics, Incomplete
    2. 2. Greedy Algorithms, Incomplete
    3. 3. Approximate Solutions, Incomplete
    4. Quiz 5 (1 Question), Incomplete
    5. Lab 5, Incomplete
6. Part 6: Combinatorial Game Theory, Winning Strategies, Incomplete section
    1. 1. Combinatorial Game Theory, Incomplete
    2. 2. Computing Winning Positions in a Game, Incomplete
    3. Quiz 6 (1 Question), Incomplete
    4. Lab 6, Incomplete

Graph Search, Shortest Paths, and Data Structures Course by Stanford University

This is one of 4 in the Algorithms Specialization
The primary topics in this part of the specialization are: data structures (heaps, balanced search trees, hash tables, bloom filters), graph primitives (applications of breadth-first and depth-first search, connectivity, shortest paths), and their applications (ranging from deduplication to social network analysis).

1) ***WEEK 1***

Breadth-first and depth-first search; computing strong components; applications.
Graded: Programming Assignment #1

2) ***WEEK 2***

Dijkstra's shortest-path algorithm.

3) ***WEEK 3***

Heaps; balanced binary search trees.

4) ***WEEK 4***

Hashing; bloom filters.